Computer Science - Graduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (22 marks)

1. Which of the following sentence is FALSE regarding regression?
   (a) It may be used for interpretation.
   (b) It is used for prediction.
   (c) It relates inputs to outputs.
   (d) It discovers causal relationships
2. Consider the Bayesian network given below. How many independent parameters would we need
   (a) 8
   (b) 15
   (c) 10
   (d) 3
   (e) 12
3. If N is the number of instances in the training dataset, nearest neighbors has a classification run t
   (a) O(N!)
   (b) O(N log N)
   (c) O(N^2)
   (d) O(N)
   (e) O(1)
4. Which of the following pairs of 8-bit, two's-complement numbers will result in overflow when the n
   (a) 11110000, 00001111
   (b) 01111111, 10000000
   (c) 00000001, 10000000
   (d) 10000001, 10101010
   (e) 11111111, 00000001
5. Which of the following has the greatest potential for compromising a user's personal privacy?
   (a) A group of cookies stored by the user's Web browser
   (b) The Wi-Fi network the user is connected to
   (c) The user's search engine history
   (d) The Internet Protocol (IP) address of the user's computer

(e) The operating system of the user's computer

6. Which of the following is true of mutation-based fuzzing?
   (a) It only works for network-based fuzzing, not file-based fuzzing
   (b) It modifies the target program's code to test its robustness
   (c) It requires an initial error-free program to start the process
   (d) It uses a random number generator to create new inputs
   (e) It uses a fixed set of inputs to test the program

7. Calculate the Hamming pairwise distances and determine the minimum Hamming distance amor
   (a) 3
   (b) 6
   (c) 10
   (d) 7
   (e) 8

8. What's the maximum number of edges in a simple triangle free planar graph with 30 vertices?
   (a) 69
   (b) 60
   (c) 63
   (d) 84
   (e) 45

9. How many trees are there on n (n > 1) labeled vertices with no vertices of degree 1 or 2?
   (a) 2n
   (b) 2^(n-1)
   (c) 1
   (d) 2
   (e) n + 1

10. Of the following, which gives the best upper bound for the value of f(N) where f is a solution to the
   (a) O(N log(log N))
   (b) O(N^2)
   (c) O(N)
   (d) O(log N)
   (e) O((log N)^2)

11. When does a buffer overflow occur, generally speaking?
    (a) when copying a buffer from the stack to the heap
    (b) when writing to a pointer that has been freed
    (c) when the program runs out of memory
    (d) when a pointer is used to access memory not allocated to it
    (e) when a buffer is not initialized before being used

True / False (8 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Statement 1| L2 regularization of linear models tends to ma
2. True or False: The correct answer to 'Can a stream cipher have perfect secrecy?' is 'No, since th
3. True or False: The correct answer to 'Neural networks:' is 'Always require large amounts of data t
4. True or False: The correct answer to 'What would you do in PCA to get the same projection as S

Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check if all the elements in tuple have same data type or not.

End of Paper